Chapter 2
Extending the Scope of the Theory of Knowledge

Anthonie W.M. Meijers and Peter A. Kroes

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1 Introduction

The invention of the light bulb is without any doubt one of the major inventions of the nineteenth century. It changed life profoundly in that it made human activities independent of the natural light circumstances on a hitherto unknown scale. The idea of electric light goes back to Humphrey Davy. He discovered that an electric arc between two poles could produce light. This phenomenon, however, was rather unpractical for home applications and could not be an alternative to existing gas light. New ideas were explored on the basis of the theoretical work on heat production in resistant conductors. An electric current could heat a conductor to such an extent that it would glow white-hot. Joseph Wilson Swan was the first to construct an electric light bulb on these physical principles. It contained a thin strip of material (filament) in an oxygen-free environment to prevent the strip from burning. An electric current heated the strip. He had, however, serious trouble maintaining the vacuum in the light bulb. Thomas Edison was able to solve this problem and at the same time to construct stable enough carbon wire that could function as filament. On October 21, 1879, he illuminated an electric lamp that glowed continuously for 40 h. That was an enormous achievement in those days.

US Patent 223,898, filed by Edison in November 1879, is about the design and manufacturing of electric lamps. The patent describes the claim Edison makes as follows:

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- *Structural knowledge*. By structural knowledge, we mean knowledge of the physical/chemical properties of the artefact. A technical artefact is an object with a certain function, and this function is to a large extent realized by its physical structure. Knowledge of this physical structure, that is, of the artefact as a physical object, is of prime importance for engineering practice. A structural description of an artefact is a description of the thing in the language of physics (and chemistry). It is a description of its static properties, such as its geometry, its mass and the composition of the materials used, and its dynamic properties, such as its mechanical, thermodynamical, electromagnetic and chemical behaviours. In Edison's electric lamp patent, a description of these properties is to be found throughout the specification, for example, in phrases such as 'The current is conducted into the vacuum-bulb through platinum wires sealed into the glass' and in the drawing, which gives the geometrical layout of the lamp.
- *Functional knowledge*. Looking at Edison's lamp from a functional point of view, the central question is the following: What is the function of this object and how does this thing work? The answer is a functional description of the thing, which contains a characterization of the overall function and of the sub-functions of its parts. This description represents the thing as a technological artefact and details how it is designed, how its structure is related to its function, how it behaves as a technological artefact and how it should be used. In Edison's patent functional descriptions are to be found throughout the specification in phrases such as '*h h* [referring to the drawing] are the clamps, which serve to connect the platina wires, cemented in the carbon, with the leading-wires *x x*'.[5]
- *Prescriptive knowledge*. Technical artefacts are not simply given, but designed and made by engineers. What kind of knowledge is required to make them? Making things is an essential aspect of technology and the knowledge involved in making things an essential component of technological knowledge. It consists, among others, of procedures. A procedure is a series of prescribed actions that will lead to a desired result: the produced artefact. Procedures also play a role in the design, for example, when optimization procedures are used, and in the operation of an artefact (as in 'switch off the power before opening the case'). Prescriptive knowledge is also contained in Edison's patent; he explicitly writes: 'The invention further consists in the method of manufacturing carbon conductors of high resistance, so as to be suitable for giving light by incandescence, and in the manner of securing perfect contact between the metallic conductors or leading-wires and the carbon conductor.'
- *Design knowledge*. Just as making things is an essential aspect of technology, so designing things is. The knowledge involved in designing consists of, among others, design concepts and operational principles, as we have seen before.

[5] It is important to note that in engineering practice structural and functional descriptions are often mixed; Edison's description of his invention is a clear example of this. This does not mean, however, that purely structural and functional descriptions play no role in engineering practice; shop floor drawings with specifications of the materials to be used are an example of purely structural descriptions and functional decompositions of complex artefacts, an example of purely functional descriptions.

More generally, design knowledge is knowledge of how to realize functions by inventing and making physical/chemical structures for users. Apparently, this aspect of technical knowledge was of paramount importance to Edison; the proud opening sentence of his patent reads: 'Be it known that I, THOMAS ALVA EDISON, of Menlo Park, in the State of New Jersey, United States of America, have invented an Improvement of Electric Lamps, and in the method of manufacturing the same….'

- *Know-how*. The design, development and making of technical things require not only knowledge of facts but also know-how in the sense of skills. This is more or less self-evident if we take into consideration the prescriptive nature of much of engineering knowledge. The adequate performance of prescriptions often requires a lot of skills. Obviously, this kind of knowledge is not explicitly contained in Edison's patent, but is implicitly presupposed when he writes: 'that carbon filaments may be made by a combination of tar and lamp-black, the latter being previously ignited in a closed crucible for several hours and afterward moistened and kneaded until it assumes the consistency of thick putty. Small pieces of this material may be rolled out in the form of a wire as small as seven one-thousands of a inch in diameter and over a foot in length …'. These are prescriptions, but clearly the adequate performance of these directions requires know-how in the form of skill.

The taxonomy suggests that technological knowledge involves at least knowledge about (physical/chemical) *structures*, *functions* and *actions* and that it may be of a *descriptive* and *prescriptive* nature. With regard to actions, technological knowledge is often of a prescriptive nature: it tells how to achieve certain ends through actions, which involves *know-how*. Normativity is also involved in the evaluative claims that play an important role in engineering practice: objects may malfunction, one series of actions may be better to realize an aim than another, and one object or design may be better in fulfilling a function than another.[6]

5 Will the Justified True Belief Account Work?

Up till now, epistemology has focused almost exclusively on the justified true belief approach, which is specifically appropriate for the first type of knowledge we have distinguished: *structural knowledge*. The standard tripartite account of (propositional) knowledge holds that agent A knows that *p* if and only if:

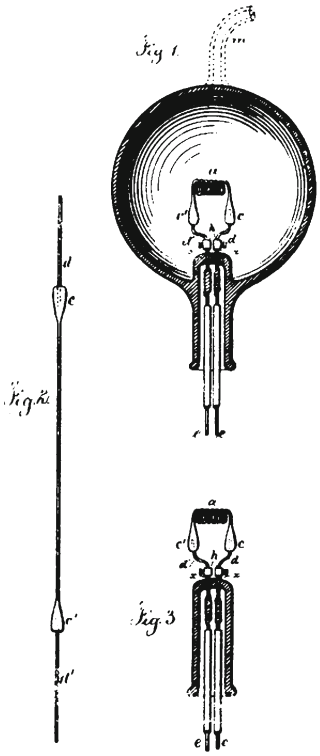
1. *p* (i.e. *p* is true).
2. A believes *p*.
3. A's belief that *p* is justified.

[6] We will not discuss extensively the kinds of knowledge involved in the *use* of technical artefacts. At first sight, it appears that this will not add any new types of knowledge to the ones mentioned so far. Clearly, prescriptive knowledge is important in this context (cf. the user manual) and know-how.

Pour information

T. A. EDISON.
Electric-Lamp.

No. 223,898. Patented Jan. 27, 1880.



Witnesses
Chas H. Smith
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Inventor
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Atty.

UNITED STATES PATENT OFFICE.

THOMAS A. EDISON, OF MENLO PARK, NEW JERSEY

ELECTRIC LAMP.

SPECIFICATION forming part of Letters Patent No. 223,898, dated January 27, 1880.

Application filed November 4, 1879.

To all whom it may concern:

Be it known that I, THOMAS ALVA EDISON, of Menlo Park, in the State of New Jersey, United States of America, have invented an Improvement in Electric Lamps, and in the method of manufacturing the same, (Case No. 186,) of which the following is a specification.

The object of this invention is to produce electric lamps giving light by incandescence, which lamps shall have high resistance, so as to allow of the practical subdivision of the electric light.

The invention consists in a light-giving body of carbon wire or sheets coiled or arranged in such a manner as to offer great resistance to the passage of the electric current, and at the same time present but a slight surface from which radiation can take place.

The invention further consists in placing such burner of great resistance in a nearly-perfect vacuum, to prevent oxidation and injury to the conductor by the atmosphere. The current is conducted into the vacuum-bulb through platina wires sealed into the glass.

The invention further consists in the method of manufacturing carbon conductors of high resistance, so as to be suitable for giving light by incandescence, and in the manner of securing perfect contact between the metallic conductors or leading-wires and the carbon conductor.

Heretofore light by incandescence has been obtained from rods of carbon of one to four ohms resistance, placed in closed vessels, in which the atmospheric air has been replaced by gases that do not combine chemically with the carbon. The vessel holding the burner has been composed of glass cemented to a metallic base. The connection between the leading wires and the carbon has been obtained by clamping the carbon to the metal. The leading-wires have always been large, so that their resistance shall be many times less than the burner, and, in general, the attempts of previous persons have been to reduce the resistance of the carbon rod. The disadvantages of following this practice are, that a lamp having but one to four ohms resistance cannot be worked in great numbers in multiple arc without the employment of main conductors of enormous dimensions; that, owing to the low resistance of the lamp, the leading-wires must be of large dimensions and good conductors, and a glass globe cannot be kept tight at the place where the wires pass in and are cemented; hence the carbon is consumed, because there must be almost a perfect vacuum to render the carbon stable, especially when such carbon is small in mass and high in electrical resistance.

The use of a gas in the receiver at the atmospheric pressure, although not attacking the carbon, serves to destroy it in time by "air-washing," or the attrition produced by the rapid passage of the air over the slightly-coherent highly-heated surface of the carbon. I have reversed this practice. I have discovered that even a cotton thread properly carbonized and placed in a sealed glass bulb exhausted to one-millionth of an atmosphere offers from one hundred to five hundred ohms resistance to the passage of the current, and that it is absolutely stable at very high temperatures; that if the thread be coiled as a spiral and carbonized, or if any fibrous vegetable substance which will leave a carbon residue after heating in a closed chamber be so coiled, as much as two thousand ohms resistance may be obtained without presenting a radiating surface greater than three-sixteenths of an inch; that if such fibrous material be rubbed with a plastic composed of lamp-black and tar, its resistance may be made high or low, according to the amount of lamp-black placed upon it; that carbon filaments may be made by a combination of tar and lamp-black, the latter being previously ignited in a closed crucible for several hours and afterward moistened and kneaded until it assumes the consistency of thick putty. Small pieces of this material may be rolled out in the form of wire as small as seven one-thousandths of a inch in diameter and over a foot in length, and the same may be coated with a non-conducting non-carbonizing substance and wound on a bobbin, or as a spiral, and the tar carbonized in a closed chamber by subjecting it to high heat, the spiral after carbonization retaining its form.

All these forms are fragile and cannot be clamped to the leading wires with sufficient force to insure good contact and prevent heating. I have discovered that if platinum wires are used and the plastic lamp-black and tar material be molded around it in the act of carbonization there is an intimate union by com-

material be molded around it in the act of carbonization there is an intimate union by combination and by pressure between the carbon and platina, and nearly perfect contact is obtained without the necessity of clamps; hence the burner and the leading-wires are connected to the carbon ready to be placed in the vacuum-bulb.

When fibrous material is used the plastic lamp-black and tar are used to secure it to the platina before carbonizing.

By using the carbon wire of such high resistance I am enabled to use fine platinum wires for leading-wires, as they will have a small resistance compared to the burner, and hence will not heat and crack the sealed vacuum-bulb. Platina can only be used, as its expansion is nearly the same as that of glass.

By using a considerable length of carbon wire and coiling it the exterior, which is only a small portion of its entire surface, will form the principal radiating surface; hence I am able to raise the specific heat of the whole of the carbon, and thus prevent the rapid reception and disappearance of the light, which on a plain wire is prejudicial, as it shows the least unsteadiness of the current by the flickering of the light; but if the current is steady the defect does not show.

I have carbonized and used cotton and linen thread, wood splints, papers coiled in various ways, also lamp black, plumbago, and carbon in various forms, mixed with tar and kneaded so that the same may be rolled out into wires of various lengths and diameters. Each wire, however, is to be uniform in size throughout.

If the carbon thread is liable to be distorted during carbonization it is to be coiled between a helix of copper wire. The ends of the carbon or filament are secured to the platina leading-wires by plastic carbonizable material, and the whole placed in the carbonizing-chamber. The copper, which has served to prevent distortion of the carbon thread, is afterward eaten away by nitric acid, and the spiral soaked in water, and then dried and placed on the glass holder, and a glass bulb blown over the whole, with a leading-tube for exhaustion by a mercury-pump. This tube, when a high vacuum has been reached, is hermetically sealed.

With substances which are not greatly distorted in carbonizing, they may be coated with a non-conducting non-carbonizable substance, which allows one coil or turn of the carbon to rest upon and be supported by the other.

In the drawings, Figure 1 shows the lamp sectionally. *a* is the carbon spiral or thread. *c c'* are the thickened ends of the spiral, formed of the plastic compound of lamp-black and tar. *d d'* are the platina wires. *h h* are the clamps, which serve to connect the platina wires, cemented in the carbon, with the leading-wires *x x*, sealed in the glass vacuum-bulb. *e e* are copper wires, connected just outside the bulb to the wires *x x*. *m* is the tube (shown by dotted lines) leading to the vacuum-pump, which, after exhaustion, is hermetically sealed and the surplus removed.

Fig. 2 represents the plastic material before being wound into a spiral.

Fig. 3 shows the spiral after carbonization, ready to have a bulb blown over it.

I claim as my invention—

1. An electric lamp for giving light by incandescence, consisting of a filament of carbon of high resistance, made as described, and secured to metallic wires, as set forth.

2. The combination of carbon filaments with a receiver made entirely of glass and conductors passing through the glass, and from which receiver the air is exhausted, for the purposes set forth.

3. A carbon filament or strip coiled and connected to electric conductors so that only a portion of the surface of such carbon conductors shall be exposed for radiating light, as set forth.

4. The method herein described of securing the platina contact-wires to the carbon filament and carbonizing of the whole in a closed chamber, substantially as set forth.

Signed by me this 1st day of November, A. D. 1879.

THOMAS A. EDISON.

Witnesses:

S. L. GRIFFIN,

JOHN F. RANDOLPH.

Order of Cancellation of Certificate of Correction of Letters Patent No. 223,898.

It is found that the following certificate has been attached to Letters Patent granted to Thomas A. Edison for improvement in "Electric Lamps," No. 223,898, dated January 27, 1880:

DEPARTMENT OF THE INTERIOR,

UNITED STATES PATENT OFFICE,

WASHINGTON, D. C., *December 18, 1883.*

In compliance with the request of the party in interest Letters Patent No. 223,898, granted January 27, 1880, to Thomas A. Edison, of Menlo Park, New Jersey, for an improvement in "Electric Lamps," is hereby limited so as to expire at the same time with the patent of the following-named, having the shortest time to run, viz.: British patent, dated November 10, 1879, No. 4,576; Canadian patent, dated November 17, 1879, No. 10,654; Belgian patent, dated November 29, 1879, No. 49,884; Italian patent, dated December 6, 1879, and French patent, dated January 20, 1880, No. 133,756.

It is hereby certified that the proper entries and corrections have been made in the files and records of the Patent Office.

This amendment is made that the United States patent may conform to the provisions of section 4887 of the Revised Statutes.

[SEAL.]

BENJ. BUTTERWORTH,
Commissioner of Patents.

Approved:
M. L. JOSLYN,
Acting Secretary of the Interior.

Now, in compliance with the request of the parties in interest, said certificate is hereby *canceled* and proper entries and corrections have been made in the files and records of the Patent Office.

In testimony whereof I have hereunto set my hand and caused the seal of the Patent Office to be affixed, this 15th day of March, 1893.

W. E. SIMONDS,
Commissioner of Patents.

Approved:
CYRUS BUSSEY,
Assistant Secretary of the Interior.

Correction in Letters Patent No. 223,898.

DEPARTMENT OF THE INTERIOR,

UNITED STATES PATENT OFFICE,

WASHINGTON, D. C., *December 18, 1883.*

In compliance with the request of the party in interest, Letters Patent No. 223,898, granted January 27, 1880, to Thomas A. Edison, of Menlo Park, New Jersey, for an improvement in "Electric-Lamps," is hereby limited so as to expire at the same time with the patent of the following named, having the shortest time to run, viz: British Patent dated November 10, 1879, No. 4,576; Canadian Patent dated November 17, 1879, No. 10,654; Belgian Patent dated November 29, 1879, No. 49,884; Italian Patent dated December 6, 1879; and French Patent dated January 20, 1880, No. 133,756;

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BENJ. BUTTERWORTH,
Commissioner of Patents.

Approved:
M. L. JOSLYN,
Acting Secretary of the Interior.

References

Audi, R. 1998. *Epistemology; a contemporary introduction to the theory of knowledge*. New York: Routledge.

Dancy, J. 1985. *Introduction to contemporary epistemology*. Oxford: Basil Blackwell Ltd.

Dewey, J. 1929. The quest for certainty; a study of the relation of knowledge and action. In John Dewey the Later Works, 1925–1953: 1929/the quest for certainty by Jo Ann Boydston (Editor), Stephen Toulmin (introduction) and John Dewey.

Houkes, W.N. 2009. The nature of technological knowledge. In A.W.M. Meijers (ed.) 2009.

Lemos, N. 2007. *An introduction to the theory of knowledge*. Cambridge: Cambridge University Press. 232 pp.

Martin, A.L. (ed.). 1998. *Epistemology: The big questions*. Oxford: Basil Blackwell.

McAllister, J.W. 1997. Laws of nature, natural history, and the description of the world. *International Studies in the Philosophy of Science* 11: 245–258.

Meijers, A.W.M. (ed.). 2009. *Philosophy of technology and engineering sciences*, 1st edn. (*Handbook philosophy of science*), vol. 9, 1453 pp. Amsterdam: North Holland

Mitcham, C. 1994. *Thinking through technology; The path between engineering and philosophy*. Chicago: University of Chicago Press (Chapter 8: "Types of technology as knowledge").

Morton, A. 1997. *A guide through the theory of knowledge*. Oxford: Basil Blackwell.

Mumford, S. 1998. *Dispositions*. Oxford: Oxford University Press.

Polanyi, M. 1978. *Personal knowledge*. London: Routledge (1958, 1st edn.).

Pollock, J.L. 1986. *Contemporary theories of knowledge*. Totowa: Rowman and Littlefield.

Pryor, J. 2001. Highlights of recent epistemology. *British Journal for the Philosophy of Science* 52: 95–124.

Searle, J.R. 1995. *The construction of social reality*. New York: The Free Press.

Sosa, E., and J. Kim (eds.). 2000. *Epistemology; an anthology*. Oxford: Basil Blackwell.

Stanley, J., and T. Williamson. 2001. Knowing how. *The Journal of Philosophy* 97: 411–444.

Vincenti, W.G. 1990. *What engineers know and how they know it; Analytical studies from aeronautical history*. Baltimore: John Hopkins University Press (Chapter 7: "The Anatomy of Engineering Design Knowledge").